

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 9: Smoke & Fire Detection and Protection Systems

Syllabus Source: CAE Oxford (pp. 281-308) | Pass Target: >= 90% AviationExam

1. FIRE CLASSIFICATIONS & EXTINGUISHING AGENTS

Aircraft fires present an immediate mortal threat in flight. Extinguishing agents must extinguish fire rapidly without generating toxic byproducts or causing electrical short circuits:

Fire Class	Combustible Material Type	Authorized Aeronautical Extinguishing Agent
Class A	Solid combustibles (wood, paper, fabrics, upholstery).	Water with antifreeze, or Halon followed by water dousing to quench deep embers.
Class B	Flammable liquids (aviation kerosene, hydraulic oil, lubricating grease).	Halon 1211 (BCF) or Halon 1301. Smothers flame; NEVER USE WATER (causes explosive liquid spread)!
Class C	Energized electrical equipment (avionics, circuit breaker panels).	Halon 1211 / 1301 or CO ₂ . Non-conductive agent mandatory to prevent pilot electrocution and equipment corrosion.
Class D	Combustible metals (magnesium wheels, titanium engine components).	Dry powder extinguishing agents (sand, copper powder). Very difficult to extinguish; water reacts violently!

2. FIRE DETECTION SYSTEMS: CONTINUOUS THERMAL LOOPS

Engine nacelles and APU compartments use continuous loop thermal detectors routed around high-risk zones:

- **Fenwal System (Resistance):** A continuous Inconel tube packed with a ceramic thermistor core surrounding a central nickel conductor wire. As temperature rises anywhere along the loop, electrical resistance of the ceramic drops exponentially, allowing current to flow and triggering the fire alarm.
- **Kidde System (Dual Element):** Two wire conductors embedded in a thermistor material inside an Inconel tube, measuring resistance and capacitance changes.
- **Systron-Donner (Pneumatic Pressure):** Stainless steel tube sealed with a helium gas charge and a central core saturated with titanium hydride (releasing hydrogen at high temperatures). Average overheat expands the helium gas (closing a low-pressure switch); localized fire releases hydrogen gas (closing a high-pressure fire switch). If the tube is severed, helium leaks out, triggering a FAULT/INTEGRITY warning!
- **Dual-Loop Logic (AND vs OR):** Engines feature two independent loops (Loop A and Loop B). Normal logic is 'A AND B' (both loops must sense fire to trigger cockpit alarm, eliminating false warnings). If one loop fails, logic switches automatically to single loop ('A OR B').

[!] EXAM TRAP (AVIATIONEXAM) - Engine Fire Drill Sequence (The Fire Handle Pull)

Pulling an Engine Fire Shutoff Handle accomplishes five vital isolation actions mechanically and electrically: 1. Shuts the engine fuel low-pressure (LP) firewall shutoff valve; 2. Shuts the engine hydraulic supply firewall shutoff valve; 3. Closes the engine pneumatic bleed air valve; 4. De-excites the engine AC generator (trips generator breaker and field breaker); 5. ARMS the fire extinguisher bottle discharge squibs (cartridges)!

3. SMOKE DETECTION & CARGO COMPARTMENT CLASSIFICATIONS

Cargo compartments under CS-25 / FAR-25 are classified according to accessibility and fire suppression capabilities:

Class	Accessibility & Suppression	Aviation Regulations & Examples
Class A	Readily accessible to crew in flight (e.g. cockpit coat closet).	Fire easily discovered; hand fire extinguisher sufficient.
Class B	Accessible to crew in flight with sufficient access to reach any part.	Equipped with smoke/fire detection and separate hand fire extinguishing access.
Class C	INACCESSIBLE lower cargo holds on passenger airliners (standard baggage hold).	MANDATORY: 1. Separate smoke/fire detector system; 2. Built-in built-in Halon extinguishing system controlled from cockpit; 3. Means to control ventilation to starve fire of oxygen.
Class D	Obsolete class (unventilated, relied on oxygen starvation). Banned following ValuJet 592 crash!	All former Class D holds have been retrofitted to Class C standards with active fire extinguishing.
Class E	Main deck cargo cabins on ALL-CARGO freighter aircraft.	Equipped with smoke detection; crew shuts down ventilation and depressurizes cargo cabin in flight to starve fire.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Fire Detection & Dual-Shot Discharge

Rule 1: Engine fire loops use continuous-wire systems (Fenwal / Kidde / Systron-Donner): - Systron-Donner: Titanium center wire stores helium and hydrogen; core gas release triggers pressure switch. - Kidde: Thermistor ceramic lowers electrical resistance as temperature rises. Rule 2: Dual-Shot Protocol: - Discharge Bottle 1 (Halon) -> Wait 30 seconds. - If fire warning persists, discharge Bottle 2 (cross-feed discharge).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What does pulling an engine fire shutoff handle in the cockpit accomplish?

- [A]: Discharges both fire bottles immediately into the engine core
- [B]: Shuts off fuel, hydraulics, bleed air, trips the generator, and arms the fire bottle squibs
- [C]: Feather the opposite engine
- [D]: Extends the landing gear

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The fire handle isolates all fluid and electrical supplies to the engine (fuel, hydraulics, pneumatic bleed, generator) and arms the explosive squibs. Rotating the handle then discharges the extinguisher bottle.

Q2: In a dual-loop engine fire detection system (Loop A and Loop B), what is the normal alarm logic to prevent

- [A]: Loop A must detect fire, while Loop B must detect smoke
- [B]: Both Loop A AND Loop B must detect the fire condition simultaneously
- [C]: Either Loop A OR Loop B is sufficient
- [D]: Loop A tests Loop B every 30 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dual-loop systems require 'AND' logic to trigger master warning bells and lights. If both loops agree, a fire is confirmed. If only one triggers, a loop fault is annunciated unless manual override is selected.

Q3: Which cargo compartment class applies to the inaccessible lower underfloor baggage holds of modern pa

- [A]: Class A
- [B]: Class B
- [C]: Class C
- [D]: Class E

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Class C cargo holds are inaccessible in flight and must have smoke/fire detection, built-in cockpit-controlled extinguishing systems (Halon flooding and metering bottles), and ventilation shutoff controls.